

CTRL-C2000 control card

Rev E0 | 2026-10-04 | Schematics authored by Chinmoy Bhuyan | TMS320F28388D common controller for PV-P75/100/110 and DAB-D60

01 Power regulators

+24V -> LMR38020 5 V -> TPS62133 3.3 V; PG chain enables TPS62130 1.2 V and TPS7A2033 3.3 V analog (VDDIO first, then VDD and VDDA)

File: sheets/01_power.kicad_sch

02 Supervisors, reset, clock, boot

TPS3703 windows on 3.3 V and 3.3 V analog, UV supervisor on 1.2 V -> XRSn; 25 MHz CMOS clock into X1; boot straps GPIO72/GPIO84 (flash by default)

File: sheets/02_reset.kicad_sch

03 TMS320F28388D (ZWT 337-ball)

All 337 balls, pin table extracted from SPRSP14E (gen/data/f28388d_pins.csv); net per ball from the pin plan (gen/data/ctrl_c2000_pin_plan.csv)

File: sheets/03_mcu.kicad_sch

04 MCU supply decoupling

SPRSP14E 7.10.1.5: 0.1 uF per VDDIO/VDD3VFL/VDDOSC ball, 2.2 uF per VDDA ball, -22 uF total on VDD + one 56 R load, 22 uF per VREFHI (16-bit mode)

File: sheets/04_decoupling.kicad_sch

05 ADC reference, VMID, slow mux

REF5025E 2.5 V for VREFHIA-D; OPA2388 buffers VMID 1.25 V (diff-amp offset) and VBIAS 2.5 V (ID/NTC bias); TMUX1208 for 4 NTC + 4 ID; 24 V monitor; AGND star

File: sheets/05_adcref.kicad_sch

06 Fault receivers, enable, EN drivers

Cn_EN = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only while CHB_OK, MCU enable and RDY; EN drivers with readback; fail-safe FLT receivers

File: sheets/06_trip.kicad_sch

07 Cell port C1

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/07_cell1.kicad_sch

08 Cell port C2

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/08_cell2.kicad_sch

09 Cell port C3

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/09_cell3.kicad_sch

10 Cell port C4

CELL contract 2x20: 8 PWM RS-422 drivers, 3 dual-range analog inputs, RDY, ID, NTC; gate-driver supply +24V_GD through a 1.5 A eFuse (TPS26600)

File: sheets/10_cell4.kicad_sch

11 Port-board analog inputs

PORT contract 2x13: VA, VAX, VB, VBX, IA, IB, AUX1, AUX2 isolated-amplifier pairs, biased open-safe, through difference amp + RC + clamp; ID: +24V out through an eFuse

File: sheets/11_port.kicad_sch

12 SYS-IO-AUX connector

80-way SYS contract: +24 V logic, +24V_GD (channel B), CAN, SCI, SPI, I2C, DI/DO, fault latch, GATE_EN, CHB_OK, watchdog, reset, fans, Ethernet MDI + centre tap

File: sheets/12_sys.kicad_sch

13 Ethernet PHY DP83822I

MII to the CM EMAC (GPIO105-124, mux 14), own 25 MHz oscillator, 49.9 R MDI terminations to AVD; MDI pairs, LEDs and the centre-tap supply ETH_CT go to SYS

File: sheets/13_eth.kicad_sch

14 JTAG, UART, EEPROM, LEDs, SDFM

cTI-20 JTAG, CM debug UART, 24LC256 calibration EEPROM (0x57), status LEDs, ERRORSTS, 8-channel SDFM expansion header (ECO-02 / ECO-09)

File: sheets/14_debug.kicad_sch

Protection without firmware: FLT_LATCH_N -> INPUTXBAR1 -> TZ1; each cell's FAULT (INPUTXBAR2-5) and AN1 CMPSS window -> TRIP4/5/7/8; port OV (ADC PPB on VA/VB, CMPSS backup), open inputs (CMPSS low), IA/IB SC (CMPSS) and OC (PPB), P_ID (PPB) -> TRIP9 for every cell; E-stop -> INPUTXBAR6 -> TRIP6. CMPSS reference = VDAC = REF5025E; firmware nulls each channel at zero input at start-up; trip bands in the pin plan, asserted against the frozen earlier-platform snapshot gen/data/control_spec_platform1.json (D-044). EN pair = GATE_EN AND CHB_OK AND MCU enable (74LVC11A); line drivers on only with CHB_OK, MCU enable and RDY; EN readback on GPIO100-103. +24V_GD feeds only the cell eFuses. Rails: 24 V -> 5 V (LMR38020) -> 3.3 V (TPS62133); its PG enables 1.2 V (TPS62130) and 3.3 V analog (TPS7A2033), so VDDIO leads VDD and VDDA (SPRSP14E table 7-1). Three TPS3703 hold XRSn (windows on 3.3 V and 3.3 V analog, UV on 1.2 V), every tolerance corner asserted. REF5025E runs from the analog rail, so VREFHI and VDAC never exceed VDDA. Pin plan with mux positions and trip bands: gen/data/ctrl_c2000_pin_plan.csv (checked against SPRSP14E table 6-1 on every build). Pin table: gen/data/f28388d_pins.csv (from the PDF). SYS side: inputs via 74LVC244A / 74LVC1G17 (loff); I2C pull-ups and EEPROM 0x57 here; ETH_CT = PHY AVD; PORT +24V via its own eFuse.

Not bench-validated. Values marked calculated are not simulated.
Sheet 1
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

24 V -> 5 V (LMR38020, 400 KHZ)

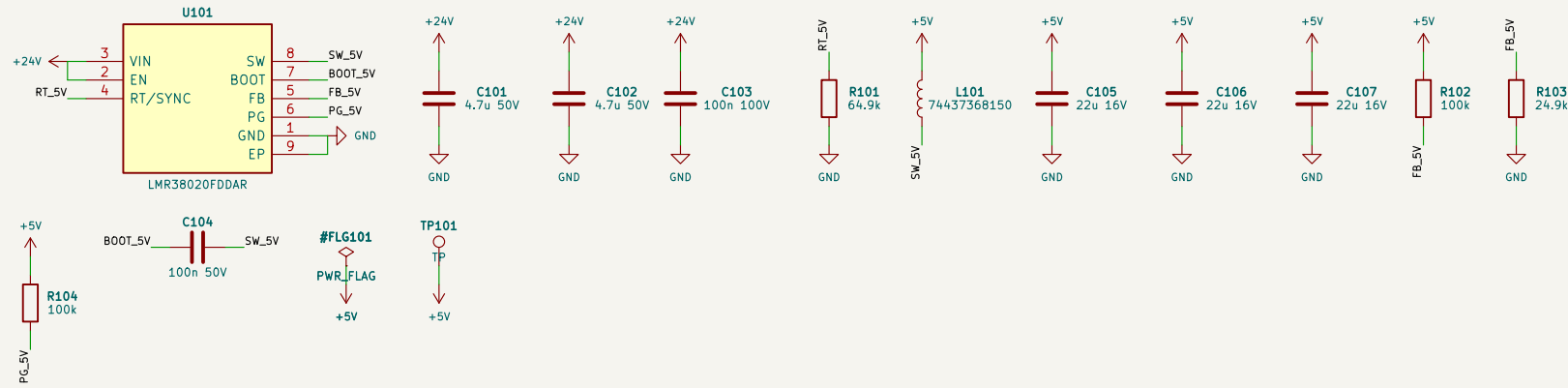
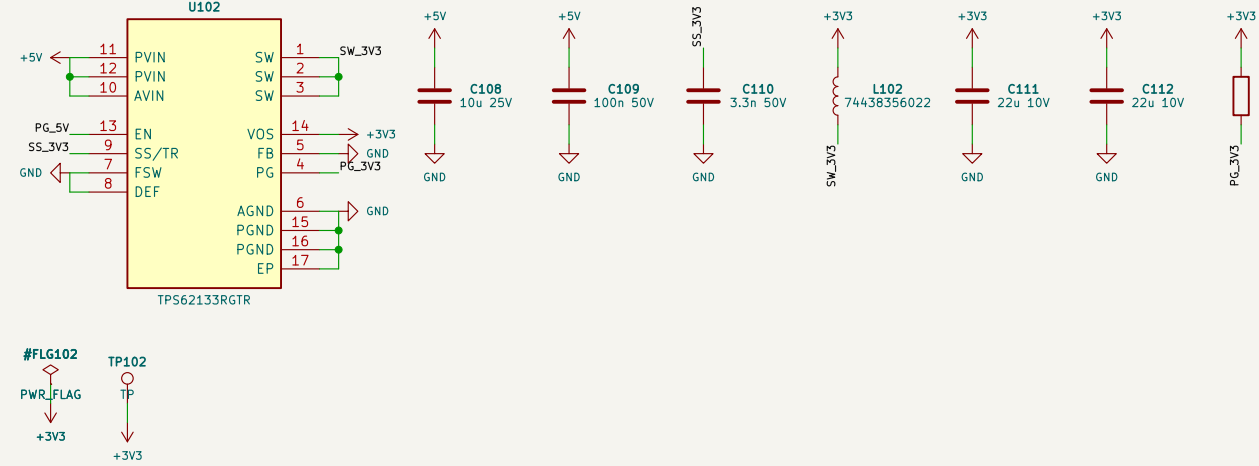


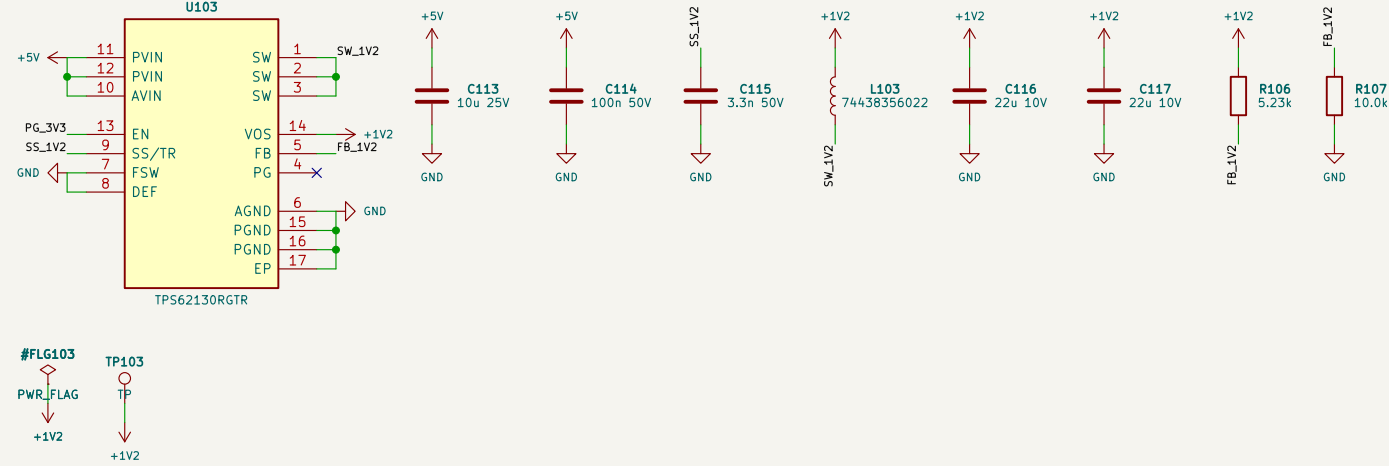
Table 9-1 row 400 kHz / 5 V / 2 A: L 15 uH, 3x22 uF, RFBT 100k / RFBB 24.9k -> 5.02 V, RT 64.9k.
Load (calculated): 3.3 V 1.3 A + 1.2 V 0.48 A + 3.3 V analog 0.1 A -> about 1.2 A at 5 V.
EN tied to VIN (datasheet allows), PG_5V enables the 3.3 V converter.
Output capacitance check (SNVSC40E, calculated); total on +5V = 3x22 + 2x10 + 1 + 0.3 = 87 uF nominal,
96 uF at +10 %; 1.45x the 66 uF design value, inside the 9.2,2.5 limit (10x design or 1000 uF) -> no extra
stability study required. Start-up: tSS 4.0 ms (8.3.9) -> 96 uF x 5.02 V / 4 ms = 0.12 A charge current,
downstream loads held off by the PG chain, << ILS-LIMIT 1.8 A min; OCP blanking 18 ms -> no hiccup.
Inductor: LMIN = 0.25 x 5 V / 400 kHz = 3.1 uH (eq. 11) < 15 uH; ripple 0.64-0.68 A at 21.6-26.4 V in
(> 10 % of 2 A); peak 1.54 A (1.62 A at L -20 %) < IHS-LIMIT 2.6 A min < ISAT 5.55 A. Bode not measured.

5 V -> 3.3 V DIGITAL (TPS62133, 3 A)



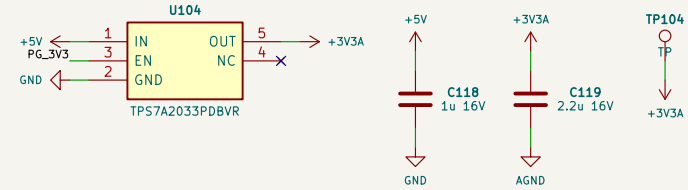
Fixed 3.3 V; FB to AGND per datasheet, FSW low (2.5 MHz), 2.2 uH, 2x22 uF (table 9-1).
SS 3.3 nF (typical application), PG_3V3 releases the 1.2 V and 3.3 V analog rails.
Worst case 3.221-3.396 V (power-save mode, SLVSAG7F p5) inside VDDIO 3.14-3.47 V.

5 V -> 1.2 V CORE (TPS62130, 3 A)



VOUT = 0.8 V x (1 + 5.23k/10.0k) = 1.218 V (eq. 6), 0.1 % 10 ppm/K; worst case 1.1872-1.2558 V
(power-save VFB 781.6-822.4 mV, FB leakage, 0.1 % line/load) inside VDD 1.14-1.26 V (CSR-01).
Enabled by PG_3V3: VDD rises after VDDIO (SPRSP14E table 7-1). VDD bulk on the decoupling sheet.

5 V -> 3.3 V ANALOG (TPS7A2033)



VDDA, op amps, reference, mux. Enabled by PG_3V3 so VDDA never leads VDDIO.
Load (calculated): 26 op-amp channels x 2.6 mA max + IDDA 15 mA + REF and its 4.2 mA load + VBIAS
+ 12 open-input bias resistors to +3V3A (16 mA) = about 110 mA of 300 mA.

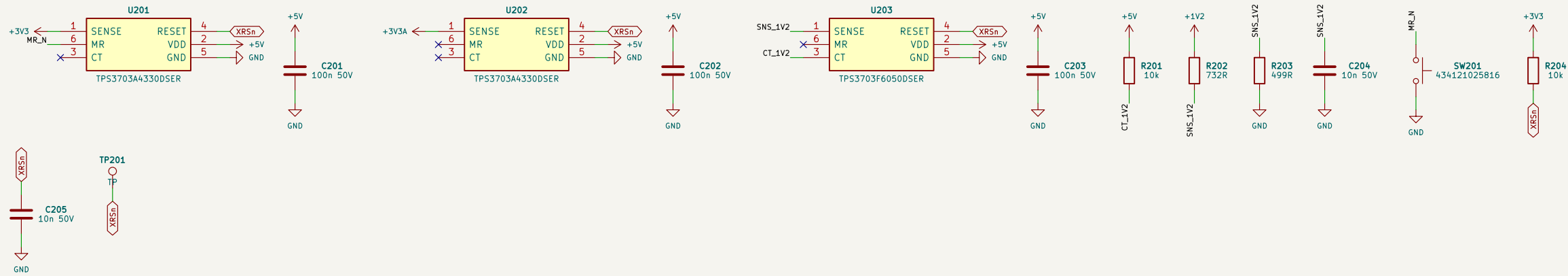
Not bench-validated. Values marked calculated are not simulated.
Sheet 2
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /01 Power regulators/
File: 01_power.kicad_sch

Title: CTRL-C2000 control card - Power regulators

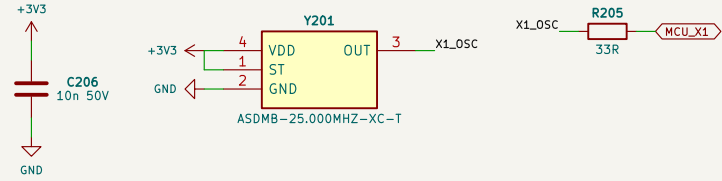
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 2/15

SUPPLY SUPERVISORS (OPEN-DRAIN RESETS WIRED TO XRSN), CSR-01



SPRSP14E 7.10.1.2.4: VDDIO and VDD need an external supervisor (the internal POR trips below range). Worst case at every corner (asserted on each build, MCU window 3.14–3.47 V / 1.14–1.26 V):
+3V3 A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2209$; OV 3.408–3.456 $> \text{regulator max } 3.3958$ V
+3V3A A4330: trips ≥ 3.1458 , releases $\leq 3.2157 < \text{regulator min } 3.2505$; OV 3.408–3.456 $> \text{regulator max } 3.3495$ V
+1V2 F6050: trips ≥ 1.1454 , releases $\leq 1.1829 < \text{regulator min } 1.1872$; regulator max 1.2558 V
1.2 V is UV-only: a $\pm 4\%$ window cannot hold the TPS62130 power-save spread; its maximum is the regulator arithmetic above. Delay: CT open 10 ms (A parts), CT to VDD via 10k 20 ms (F part). VDD = +5V so the outputs stay defined while 3.3 V collapses. XRSn pull-up 10k, 10 nF; SYS_RST_N and the JTAG RESET are open drain on the same node. Reset button on MR of the 3.3 V part (100k internal pull-up).

25 MHZ CLOCK (X1)



SPRSP14E 7.10.3.2.1: X1 from an oscillator 10–25 MHz, tr/ff ≤ 6 ns, 45–55 %; X2 open (figure 7–11). No crystal, so VSSOSC goes to board ground (table 6–12). 33 R source termination.

BOOT MODE (GPIO72 = PIN 1, GPIO84 = PIN 0)

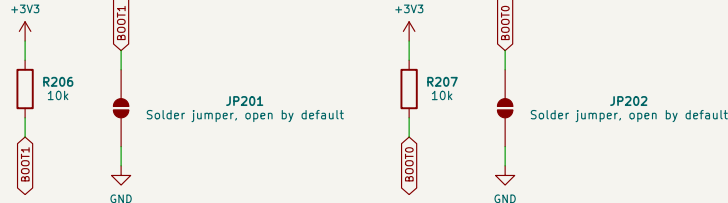
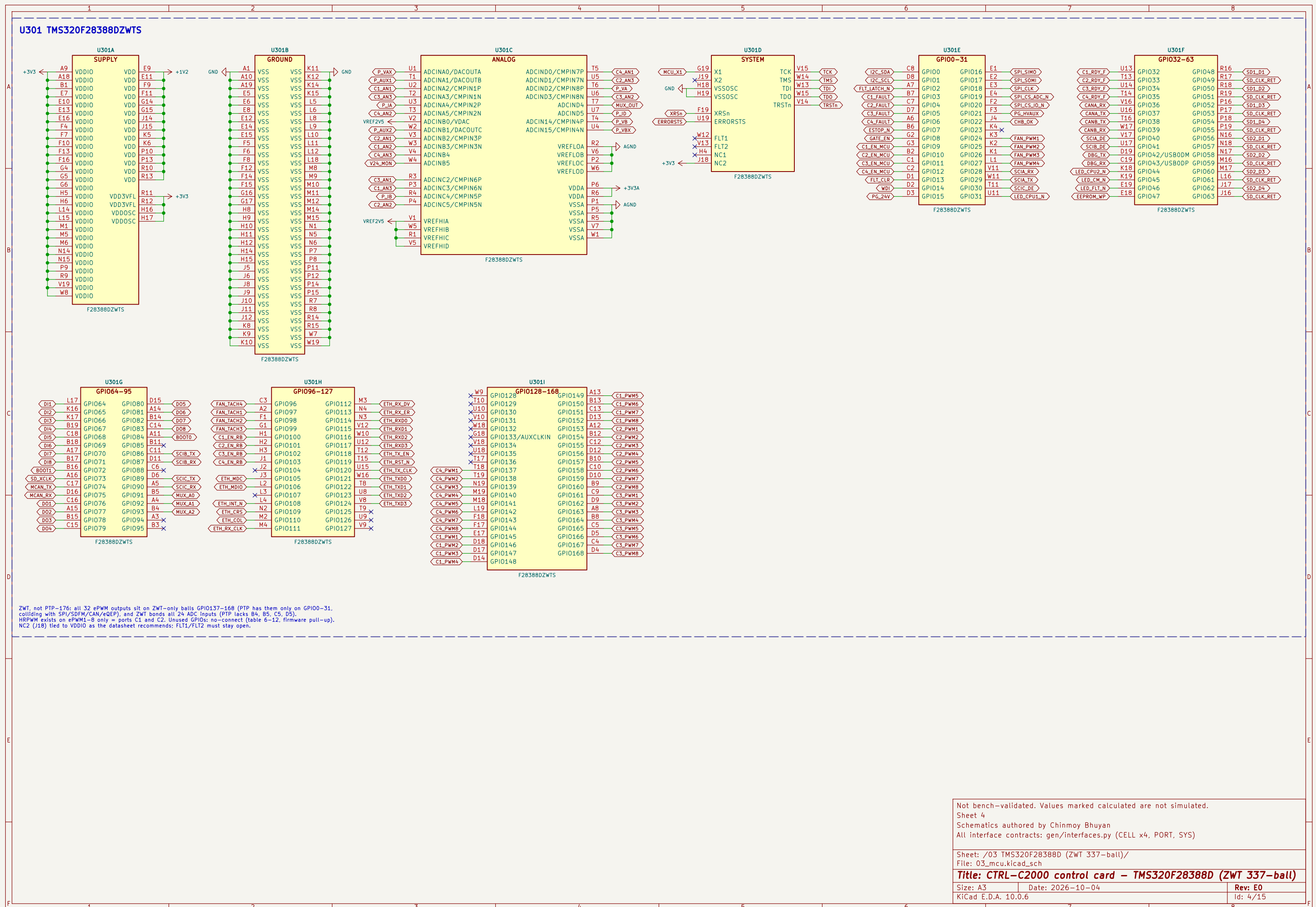
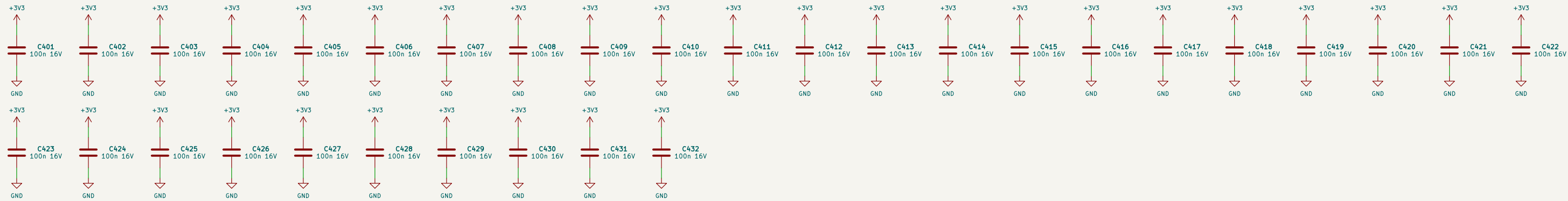


Table 8–18: 11 = flash (straps open), 01 = SCI, 10 = CAN, 00 = parallel I/O. Close JP to pull a pin low. th(boot-mode) 1.5 ms; nothing else drives these balls.

Not bench-validated. Values marked calculated are not simulated.
Sheet 3
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

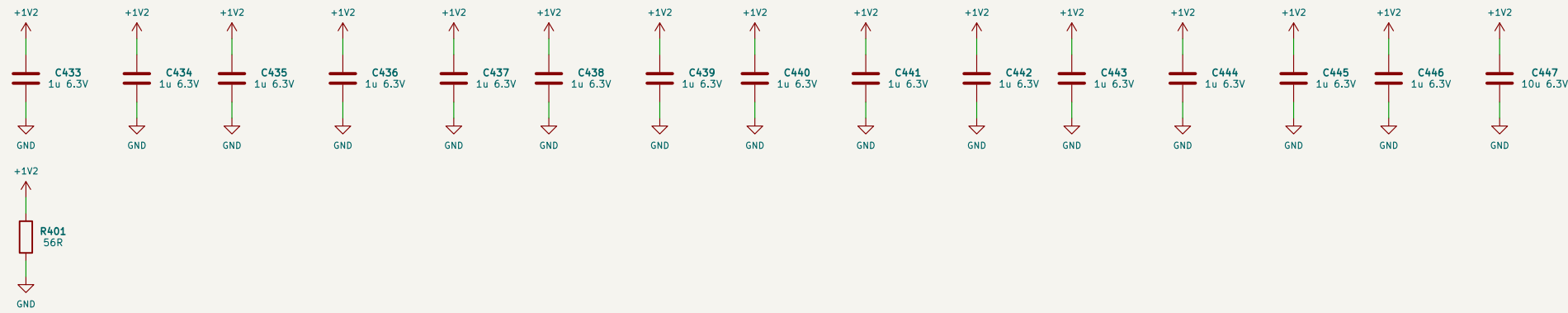


VDDIO (28 BALLS) + VDD3VFL (2) + VDDOSC (2): 0.1 uF EACH



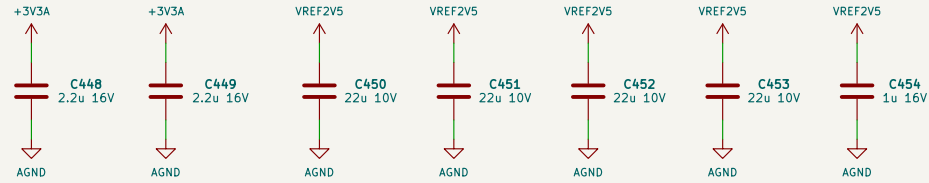
Configuration 1 of 7.10.1.3.2.

VDD 1.2 V (14 BALLS)



14 x 1 uF + 10 uF = 24 uF, 10 % parts: >= 21.6 uF (CVDD TOTAL 20 min / 22 typ. max tolerance 20 %).
56 R load consumes the VDD3VFL-to-VDD leakage current (table 6-4 note).

VDDA, VREFHI AND VDAC



VDDA 2.2 uF per ball to VSSA. VREFHIA-D: 22 uF each to VREFLO (16-bit mode).
VDAC (ADCINB0, V2) = CMPSS DAC reference: >= 1 uF at the ball (table 6-1).
All on VREF2V5: 4 x 22 + 1 + 1 uF = 90 uF, 99 uF at +10 %: inside the REF5025E 1-100 uF range (summed from the design on every build).

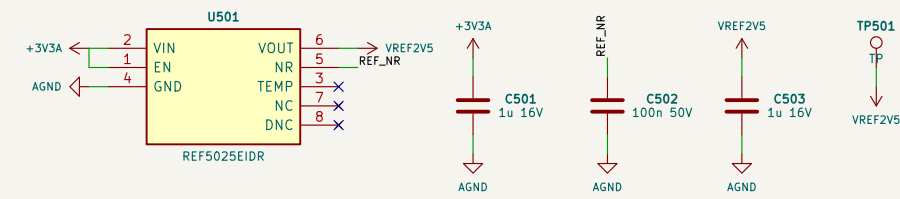
Not bench-validated. Values marked calculated are not simulated.
Sheet 5
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /04 MCU supply decoupling/
File: 04_decoupling.kicad_sch

Title: CTRL-C2000 control card - MCU supply decoupling

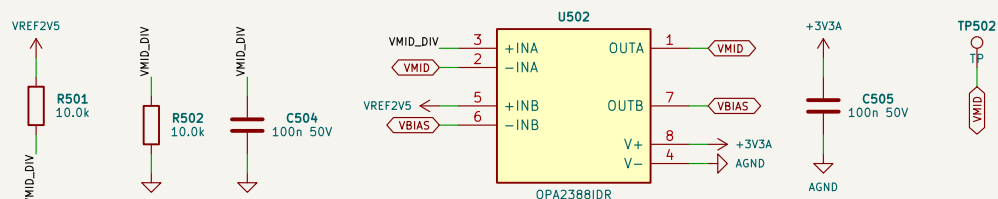
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 5/15

2.5 V ADC AND CMPSS DAC REFERENCE



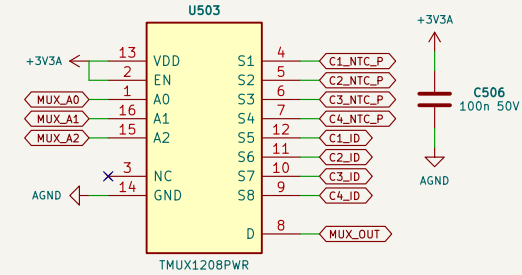
Powered from +3V3A (VIN >= VOUT + 0.2 V) so VREFHI and VDAC can never exceed VDDA (SPRSP14E 7.11.2.3.5, 7.11.3.1.4). Also the CMPSS DAC reference (VDAC, INT-05); thresholds ratiometric with the ADC.
Load 4.2 mA of +/-10 mA: VDAC 8 x 6 k (3.3 mA) + VREFHI 4 x 190 uA + VMID divider (asserted).
NR 0.1 uF class 1; 1 uF HF at VOUT; VREFHI and VDAC capacitors are on the decoupling sheet.

VMID AND VBIAS BUFFERS



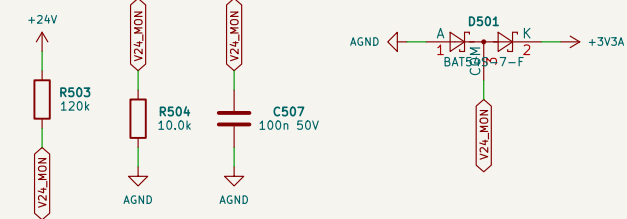
VMID = VREF/2 is ratiometric: a zero differential input reads mid-scale whatever the reference error.
VBIAS = VREF buffered, so ID and NTC readings are ratiometric and the reference itself is not loaded.

SLOW-SIGNAL MUX -> ADCIND4



S1-S4 = C1-C4 NTC, S5-S8 = C1-C4 ID (24 ADC balls cannot take 20 fast channels + 9 slow ones).
Each input has its own 100 nF on the cell sheet; the mux only switches between charged caps.

24 V MONITOR -> ADCINB5



120k/10k: 24 V reads 1.85 V, full scale 32.5 V: 100 nF filter.
+24V is live before VDDA: the BAT54S holds the ball at VDDA + VF (7.10.1.4.2).

AGND - GND STAR POINT



Single connection of the analog return to board ground (layout: under U301).

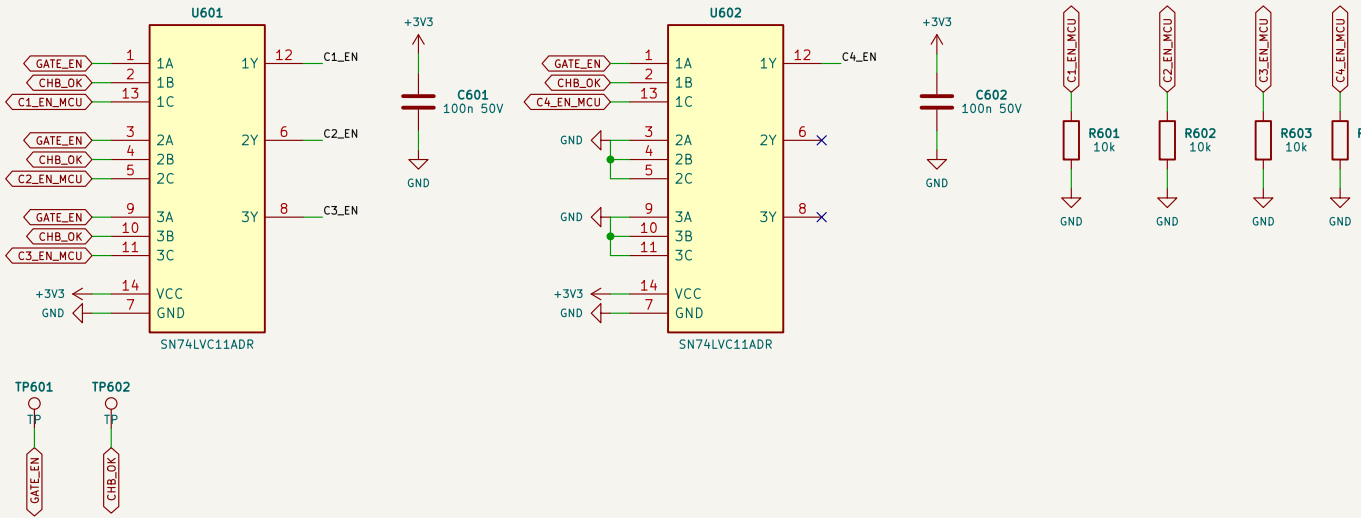
Not bench-validated. Values marked calculated are not simulated.
Sheet 6
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /05 ADC reference, VMID, slow mux/
File: 05_adcref.kicad_sch

Title: CTRL-C2000 control card - ADC reference, VMID, slow mux

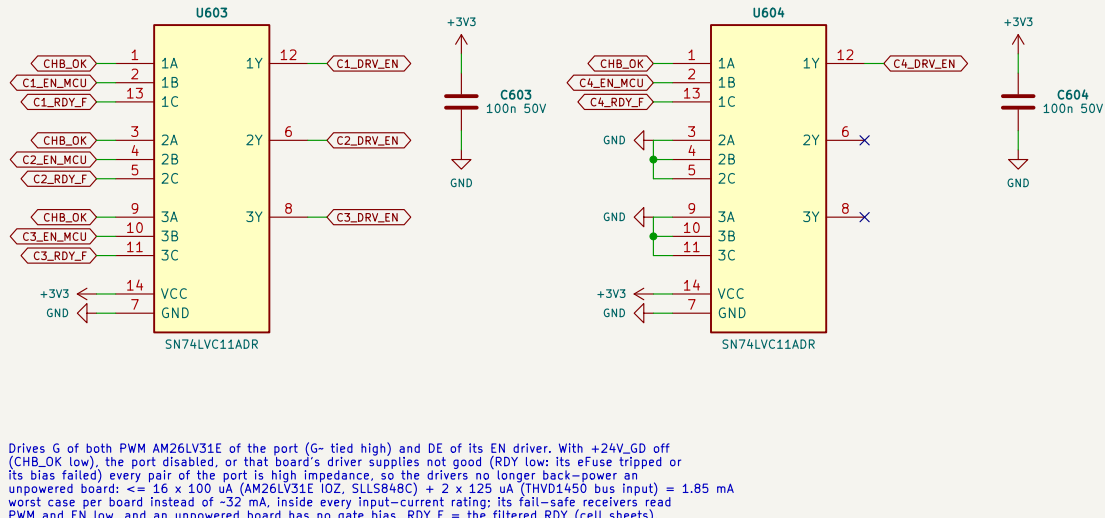
Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 6/15

HARDWARE GATE-ENABLE AND (BOTH SAFETY CHANNELS)



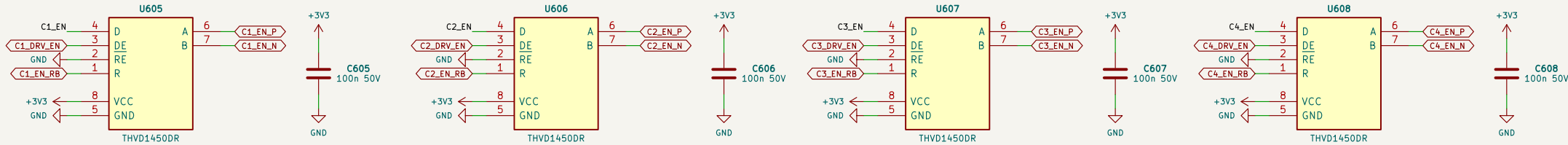
Cn_EN = GATE_EN (channel A) AND CHB_OK (channel B) AND Cn_EN_MCU, one 3-input gate per port; firmware alone can never raise an EN pair. Channel B also removes +24V_GD (gate-driver bias) on SYS. 10k pull-downs hold each MCU enable low in reset; GATE_EN and CHB_OK are pulled low on the SYS side. Spare gates: inputs to GND, outputs open.

LINE-DRIVER ENABLE CN_DRV_EN = CHB_OK AND CN_EN_MCU AND CN_RDY (INT-15)



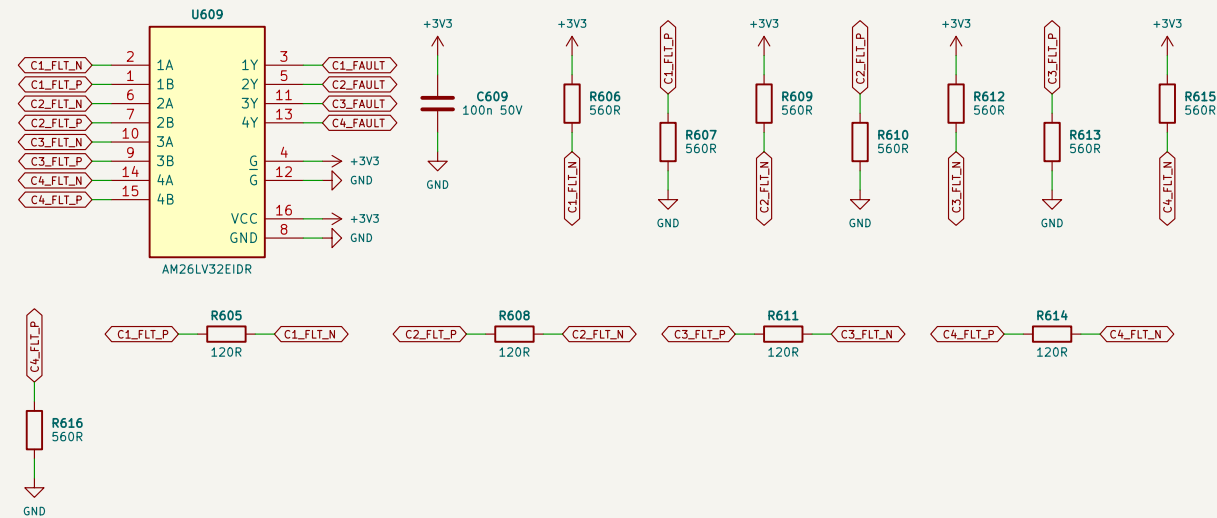
Drives G of both PWM AM26LV31E of the port (G- tied high) and DE of its EN driver. With +24V_GD off (CHB_OK low), the port disabled; or that board's driver supplies not good (RDY low: its eFuse tripped or its bias failed) every pair of the port is high impedance, so the drivers no longer back-power an unpowered board: $\leq 16 \times 100 \mu A$ (AM26LV31E IOZ, SLLS848C) + $2 \times 125 \mu A$ (THVD1450 bus input) = 1.85 mA worst case per board instead of -32 mA, inside every input-current rating: its fail-safe receivers read PWM and EN low, and an unpowered board has no gate bias. RDY_F = the filtered RDY (cell sheets).

EN LINE DRIVERS WITH READBACK (C1-C4)



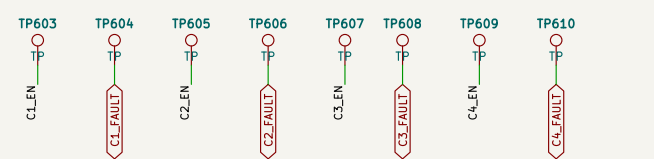
THVD1450 per port: D = Cn_EN, DE = Cn_DRV_EN, A/B = EN_P/EN_N, [VOD] ≥ 2 V into 100 R (SLLSEY3E 7.7). Receiver always on (RE- low): R = the pair's state at the CTRL end $\rightarrow$ Cn_EN_RB on a spare GPIO, so a stuck AND gate or driver is visible (INT-22: start-up and proof-test sequence in the pin plan).

FLT RECEIVERS (C1-C4), FAIL-SAFE



Pair wired swapped (A = FLT_N, B = FLT_P): Cn_FAULT is high = fault. 560R bias + 120R termination put +0.32 V across an open/unplugged pair ($\rightarrow$ VIT+ 0.2 V) $\rightarrow$ fault; the AM26LV32E rev E only has open-circuit (not terminated) fail-safe, hence the external bias. Healthy: driver -2 V $\rightarrow$ low, Cn_FAULT $\rightarrow$ INPUTXBAR2-5 $\rightarrow$ ePWM X-BAR $\rightarrow$ TRIP4/5/7/8 (no firmware in the trip path).

TEST POINTS



Gated EN (after the AND) and receiver FAULT outputs, one pair per port.

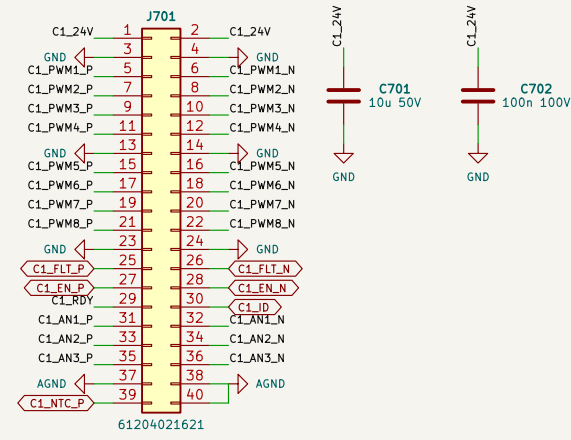
Not bench-validated. Values marked calculated are not simulated.
Sheet 7
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /06 Fault receivers, enable, EN drivers/
File: 06_trip.kicad_sch

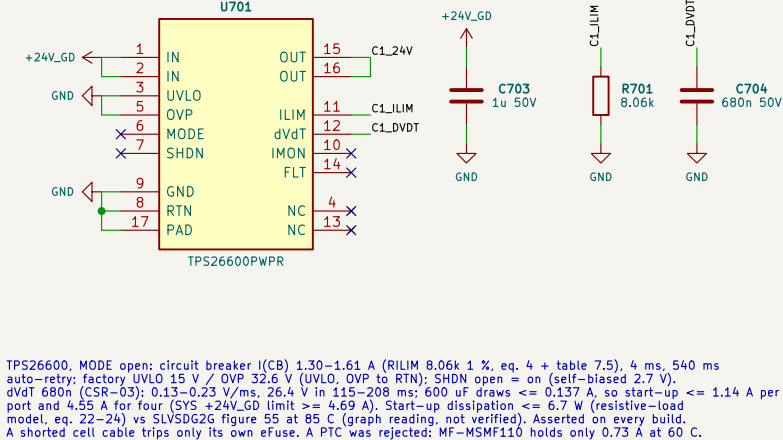
Title: CTRL-C2000 control card – Fault receivers, enable, EN drivers

Size: A3 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 7/15

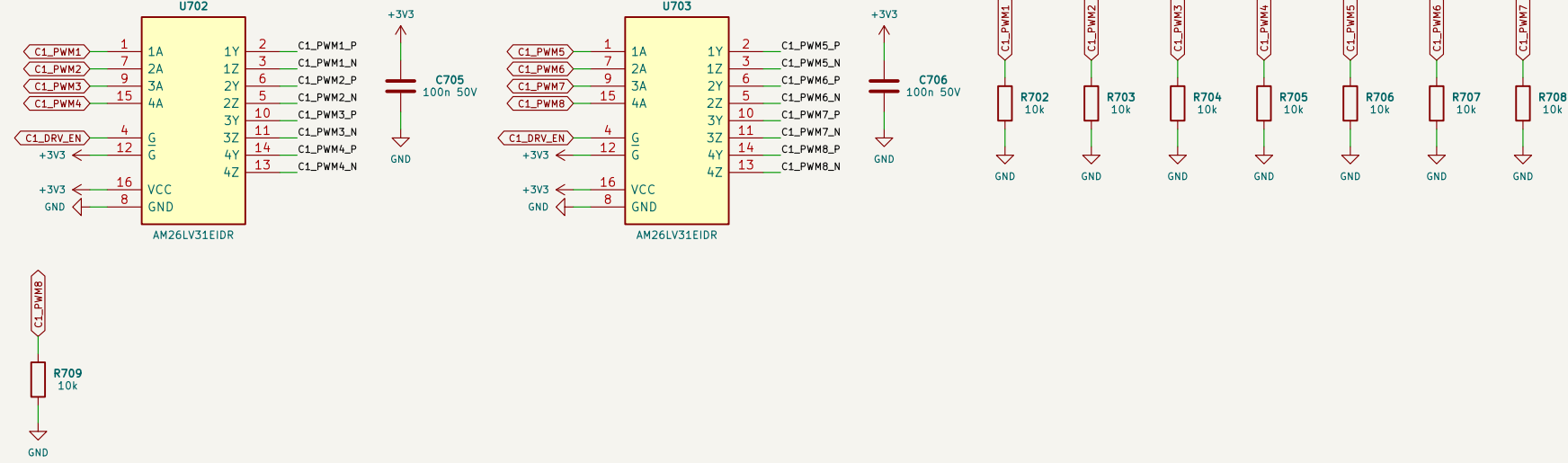
C1 CONNECTOR (CELL CONTRACT)



+24V_GD EFUSE FOR C1 (<= 1.0 A CONTINUOUS, <= 600 uF)

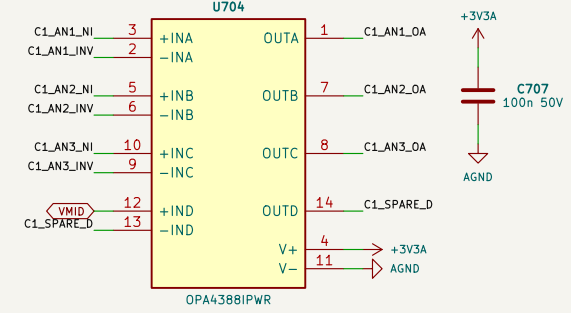


PWM LINE DRIVERS (EPWM1-4)



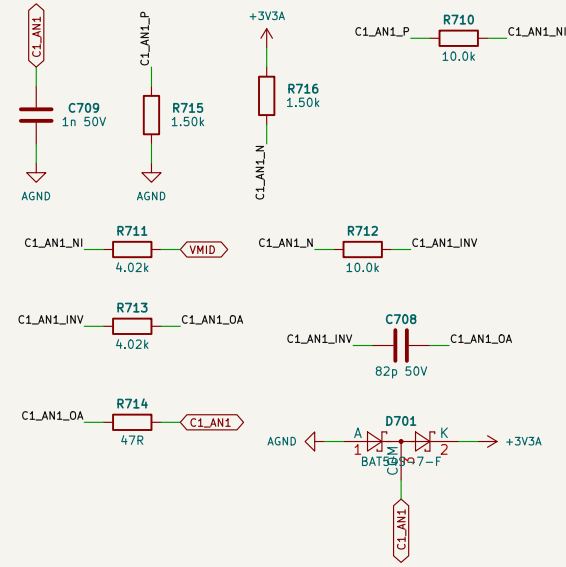
PWMx (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = C1_DRVx_EN, G=high: outputs high impedance unless C1B_0K, C1_EN_MCU and RDY are high (N1-15). ePWM1-4 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

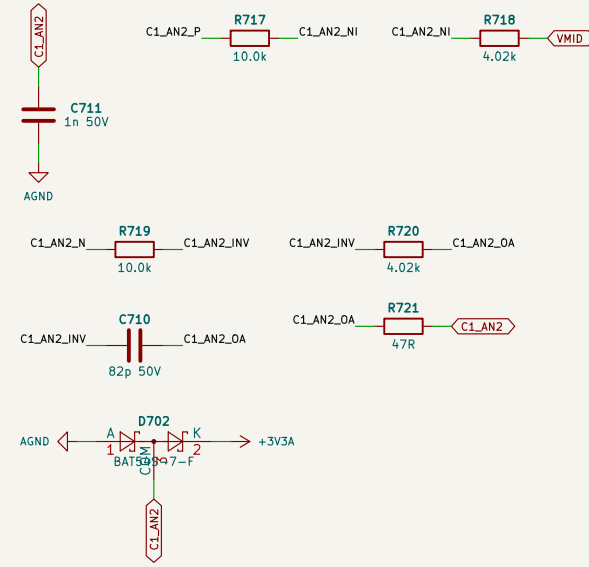


Difference amp per pair: $V_{ad} = V_{MD} 1.25 V + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 % 10 ppm/K, $G = 4.02k/10.0k = 0.402$. C1 || Rf anti-alias pole 483 kHz, 47 R + 1 nF CG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT5MS clamps the ADC mode to 3V3A/AGND. All values calculated, not simulated. Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with AN_N = AGND at the source (temperatures) -> 1.25-2.46 V, Op-amp inputs 0.9-1.75 V (RMID). Source resistance adds to the 10.0k: keep R <= 100 R (1 % gain) or calibrate it out. AN1 = inductor current (INT-12, one constant); PVCELL-25 IEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11k A -> 8.0k mV/A at the ADC. 0.0759 A/V/LSB; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it. DAB-360 (C1/C2): AN1 5.009 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMD follower.

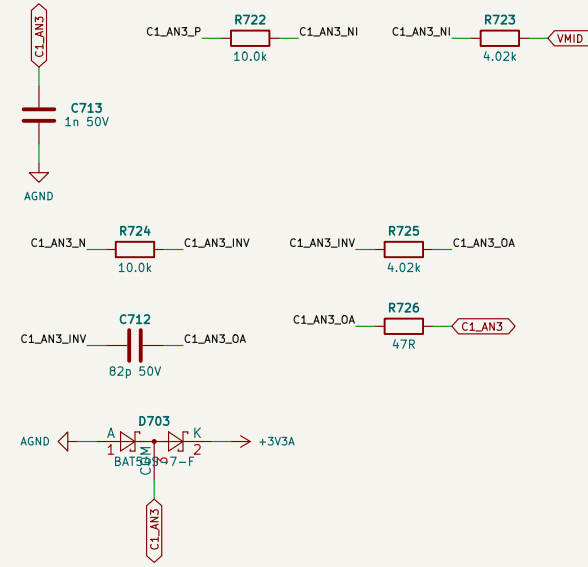
C1_AN1 -> ADCINA2 / CMPIN1P



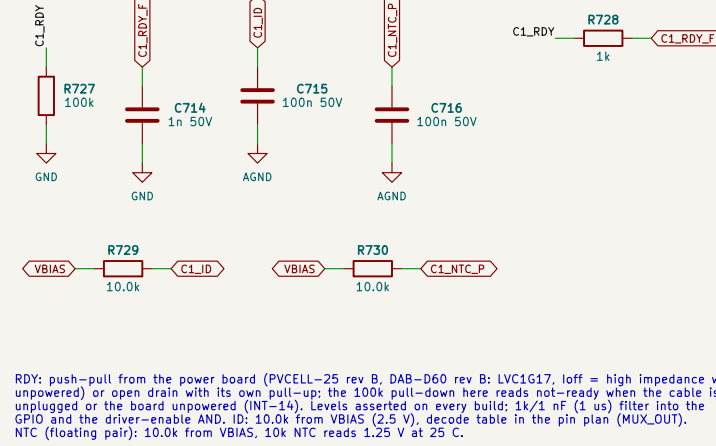
C1_AN2 -> ADCINB3



C1_AN3 -> ADCINC3



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (N1-14). Levels asserted on every build; 1k/2 nF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 8
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /07 Cell port C1/
File: 07_cell1.kicad_sch
Title: CTRL-C2000 control card - Cell port C1

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 8/15

Pinout diagram for the 40-pin J801 connector. The diagram shows two rows of pins. The left row (pins 1-20) is labeled with various control signals: C2_24V, GND, C2_PWM1_N, C2_PWM2_P, C2_PWM3_N, C2_PWM4_P, GND, C2_PWM5_P, C2_PWM6_P, C2_PWM7_P, C2_PWM8_P, C2_FILT_P, C2_LED_P, C2_LED_N, C2_RDY_P, C2_AN1_P, C2_AN2_P, C2_AN3_P, AGND, and C2_NTC_P. The right row (pins 21-40) is labeled with power and ground signals: C2_24V, GND, C2_PWM1_N, C2_PWM2_N, C2_PWM3_N, C2_PWM4_N, GND, C2_PWM5_N, C2_PWM6_N, C2_PWM7_N, C2_PWM8_N, GND, C2_FILT_N, C2_LED_N, C2_LED_P, C2_ID, C2_AIN1_N, C2_AIN2_N, C2_AIN3_N, AGND, and C2_ID. Two capacitors are shown: C801 (10u 50V) connected between pins 2 and 3, and C802 (100n 100V) connected between pins 21 and 22.

NTC_N returns to AGND here; PCT/EN pairs go to the chip sheet.

U001

+24V_GD

IN

OUT

C2_24V

C2_DVDT

ILIM

C2_ILIM

dVdt

IMON

FLT

NC

NC

NC

PAD

GND

GND

GND

C803 1u50V

R801 0.06A

C804 680n 50V

TP526600WPRP

[illegible]

PWMk (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs: ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = C2_DRV_EN, G_high: outputs high impedance unless CHB_OK, C2_PWM_MCU and RDI are high (INT-15). ePWM5-8 have high-resolution PWM: the dual-active bridge (phase shift at 100 kHz) must use ports C1 and C2, one bridge per port.

OPA4388IPWR

Pin	Signal	Pin	Signal
1	INA	7	OUTB
2	INB	8	OUTC
3	INB	9	OUTD
4	INB	10	AGND
5	INB	11	DGND
6	INB	12	AGND
7	OUTB	13	DGND
8	OUTC	14	OUTD
9	OUTD		
10	AGND		
11	DGND		
12	AGND		
13	DGND		
14	OUTD		

Pin 1: INA
Pin 2: INB
Pin 3: INB
Pin 4: INB
Pin 5: INB
Pin 6: INB
Pin 7: OUTB
Pin 8: OUTC
Pin 9: OUTD
Pin 10: AGND
Pin 11: DGND
Pin 12: AGND
Pin 13: DGND
Pin 14: OUTD

Supply: +V_{SA} (Pin 4), -V_{SA} (Pin 11)
Ground: AGND (Pin 10), DGND (Pin 13)

Capacitor: C807 (100nF) connected between pins 10 and 13.

Difference amp per pair: Vads = $125 \text{ mV} \times 0.6 \times (\text{ANP} - \text{ALN})$, Rn $10.0 \times 0.1 \times 30 \text{ pmV/K}$, $G = 0.02/(0.10 \times 0 + 0.02 \times 1)$; Rf; anti-alias pole 483 Hz, $r_{\text{in}} = r_{\text{FC}}$; ADC: analog charge bucket (ACQPS = $\pm 10 \text{ nC}$ at 12 bit).
BATS4S channels the ADC node to BVS/AGND. All values calculated, not simulated.
BVS: $V_{\text{BVS}} = 1.25 \text{ V} + 0.007 \text{ V} \times (\text{ANP} - \text{ALN})$. $V_{\text{BVS}} = 1.25 \text{ V}$ if $(\text{ANP} - \text{ALN}) > 0$. $V_{\text{BVS}} = 1.25 \text{ V}$ (ADC snap $+3.1 \text{ V}$).
 $I_{\text{BVS}} = 0$ -3.0 mA single-ended when ALN = AGND at the source (temperatures) $\rightarrow 1.25 - 2.46 \text{ V}$. Op-amp inputs over $\pm 0.75 \text{ V}$ (RIN), Source resistance adds up to the 10.0k; keep it $<= 100 \text{ k}\Omega$ (1 g gain) or calculate it. $C_{\text{BVS}} = 10 \text{ pF}$. $C_{\text{M}} < 1 \text{ pF}$ (source length) $+114 \mu\text{s} \rightarrow 8.04 \text{ mV}/\text{at}$ at the ADC, $0.0759/\Delta T^\circ\text{C}$; CMPSB: charge bucket $+91.5 \text{ fA}$. The LA UDS+ offset ($\sim 1.5 \text{ A}$) is re-zeroed by firmware at every idle (gate) off; trip bands carry only the 1.17 A left after it.
UAD: $V_{\text{UAD}} = 1.25 \text{ V} + 0.007 \text{ V} \times (\text{ANP} - \text{ALN})$. $V_{\text{UAD}} = 1.25 \text{ V}$ if $(\text{ANP} - \text{ALN}) > 0$. $V_{\text{UAD}} = 1.25 \text{ V}$ (single-ended, 49.9 R sources $+0.5 \text{ K}$ gain). Only ANI carries the open-input bias. Channel D: VMID follower.

[illegible][illegible][illegible]

Figure 1 is a schematic diagram of the power plane layout for the R827, R828, R829, and R830 components. The diagram shows the power and ground connections for each component, including the C2_RDY, C2_ID, C2_NTC_P, and C2_NTC_F capacitors. The components are connected to a common ground plane (GND) and a power plane (VBIAS). The R827 and R828 components are connected to GND and VBIAS. The R829 component is connected to GND and VBIAS. The R830 component is connected to GND and VBIAS. The C2_RDY, C2_ID, C2_NTC_P, and C2_NTC_F capacitors are connected to the power plane and ground plane.

R827 100k
GND
C2_RDY_P
C2_ID
C2_NTC_P
C2_NTC_F
R828 1k
C2_RDY_F
R829 10.0k
VBIAS
C2_ID
R830 10.0k
VBIAS
C2_NTC_P

Capacitor Values:
C814 1n 50V
C815 100n 50V
C816 100n 50V

Ground Connections:
GND
AGND

Power Connections:
VBIAS

Component Labels:
R827 100k
R828 1k
R829 10.0k
R830 10.0k
C2_RDY_P
C2_ID
C2_NTC_P
C2_NTC_F
C2_RDY_F

Notes:
R827: push-pull from the power board (PVCELL-25 rev B, DAB-D60 rev B; LVC1G17; I_{OH} = high impedance w/ unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered. (NT-14). Levels asserted on every valid 3V(1) NF (1 μ s) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS; 10k NTC reads -2.5 V at 25 $^{\circ}$ C.

RDY: push-pull from the power board (PVCCELL-25 rev B, DAB-D60 rev B; LVC1G17, Ioff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build; 1k/1 nF (1- μ s) filter into the GPIO and the driver-enable AND. ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pin): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Pinout diagram of the 40-pin J901 connector for the 6120A021621 module. The diagram shows two rows of pins. The top row (pins 1-20) includes C3_24V, GND, C3_PWM1_P, C3_PWM2_P, C3_PWM3_P, C3_PWM4_P, GND, C3_PWM5_P, C3_PWM6_P, C3_PWM7_P, C3_PWM8_P, GND, C3_FLT_P, C3_EN_P, C3_RDY, C3_AN1_P, C3_AN2_P, C3_AN3_P, AGND, and C3_NTC_P. The bottom row (pins 21-40) includes C3_24V, GND, C3_PWM2_N, C3_PWM3_N, C3_PWM4_N, GND, C3_PWM5_N, C3_PWM6_N, C3_PWM7_N, C3_PWM8_N, GND, C3_FLT_N, C3_EN_N, C3_ID, C3_AN1_N, C3_AN2_N, C3_AN3_N, AGND, and an unlabeled pin 40. Two capacitors, C901 (10u 50V) and C902 (100n 100V), are connected between C3_24V and GND on both sides.

NTC_N returns to AGND here; PCT/EN pairs go to the chip sheet.

TPS26600PWPR

TPS26600. MODE open; circuit breaker (ICB) 1.30–1.61 A (ILIM 8.06k 1%, eq. on +table 7.5), 4 ms, 540 ms auto-retry; factory UVLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN eq. on (self-biased 2.7 V).
dVdt 145–208 ms/600 uV draws <= 0.137 A; start-up <= 1.14 A per port and 4.55 A for four (SYS +24V_G0 [lim] >= 4.60 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22–24) vs SIVSDG26 figure 95 at 85 °C (graph reading, not verified). Asserted on every build. A shorted cell cable trips only if 26600 device A PTC was selected: ~~MSM110~~ holds only 0.73 A at 60 °C.

The diagram illustrates the power board circuit, featuring two AM26LV31EIDR chips (U902 and U903). The chips are connected to various PWM inputs and outputs. The inputs are labeled C3_PWM1 through C3_PWM12, and the outputs are labeled C3_DRV_EN, C3_PWM1 through C3_PWM12. The circuit includes a +3V3 supply and a GND connection. A capacitor C905 (100nF, 50V) is connected between +3V3 and GND. Another capacitor C906 (100nF, 50V) is connected between +3V3 and GND. The diagram also shows the connection of C909 (10k) to the +3V3 supply and GND.

Figure 10. Power board

PWMs (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-6), so every gate command is 0 until firmware runs. G = C3_DRV_EN, G- high: outputs high impedance unless CHB_OK, C3_EN_MCU and RDY are high (INT-15). ePWM9-12 have no high-resolution PWM; buck-boost cells only, not a DAB bridge.

Figure 1 is a schematic diagram of the OPAX38BIPWR differential amplifier circuit. The circuit is powered by +3V3A and -1V3A rails. It features two input channels, each with a 100nF capacitor (C907) connected to the +3V3A rail. The inputs are labeled C3_AN1_IN (3), C3_AN1_INH (2), C3_AN2_IN (5), C3_AN2_INH (6), C3_AN3_IN (10), and C3_AN3_INH (9). The outputs are labeled OUTA (1), OUTB (7), OUTC (8), and OUTD (14). A C3_SPARE_D (13) is also shown. The circuit is labeled OPAX38BIPWR and includes a V+ and V- supply connection.

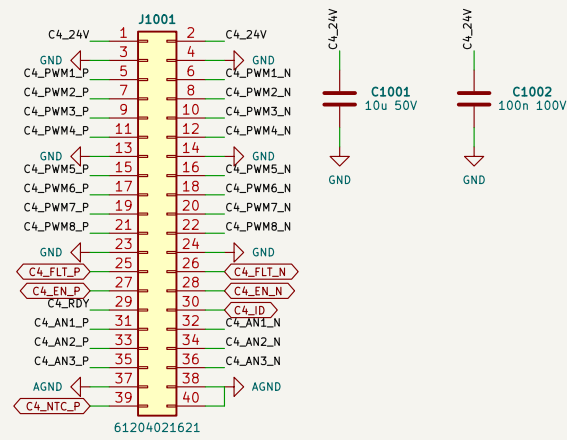
[illegible]

The schematic diagram illustrates the power supply section of the TMS320C6743 evaluation board. It features a 3.3V regulator (C911) connected to a 10k resistor (R919) and a 47R resistor (R921). The output of the regulator is connected to the VMD pin. The input of the regulator is connected to the AGND pin. The output of the regulator is also connected to the AGND pin. The input of the regulator is also connected to the AGND pin. The output of the regulator is also connected to the AGND pin.

[illegible]

RDY: push-pull from the power board (PVCILL-25 rev B, DAB-D60 rev B; LVC16V17: loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build; 1k/1 nF (1 us) filter into the GPIO and the driver-enable AND; ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX-OUT).
 NTC (floating pair): 10.0k from VBIAS; 10k NTC reads 1.25 v at 25 °C.

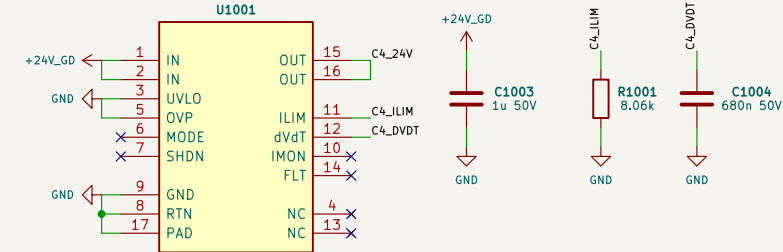
C4 CONNECTOR (CELL CONTRACT)



Pin map = Interfaces.CELL. Pins 1-2 carry +24V_GD only (gate-driver bias, channel B switched).

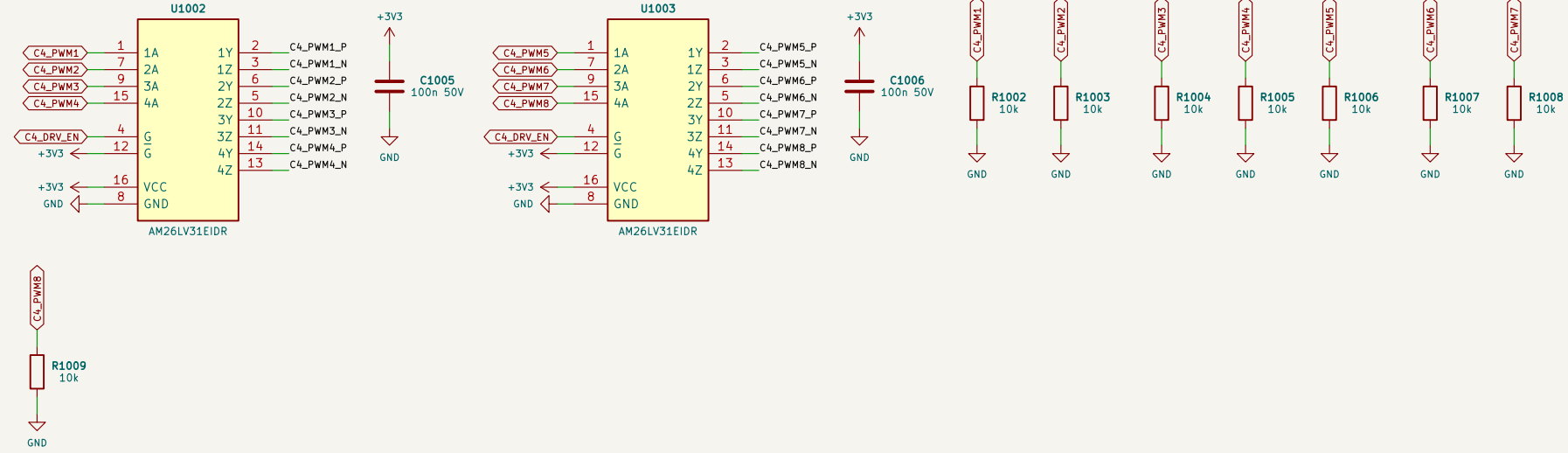
NTC_N returns to AGND here. FLV/EN gains go to the trip sheet.

+24V_GD EFUSE FOR C4 (<= 1.0 A CONTINUOUS, <= 600 uF)



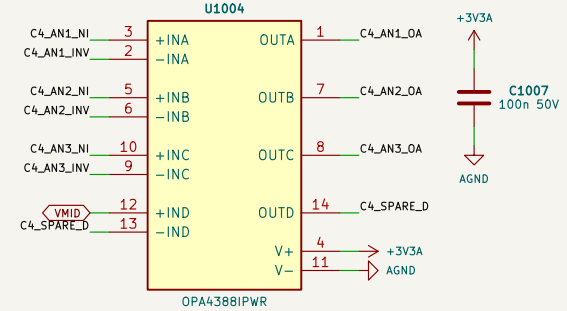
TPS26600, MODE open: circuit breaker I(CB) 1.30-1.61 A (RILIM 8.06k 1 %, eq. 4 + table 7.5), 4 ms, 540 ms auto-retry; factory UVLO 15 V / OVP 32.6 V (UVLO, OVP to RTN); SHDN open = on (self-biased 2.7 V), auto-retry; factory CSR=03; 0.13-0.23 V/ms, 26.4 V in 115-208 ms; 600 uF draws <= 0.137 A, so start-up <= 1.14 A per port and <= 55 A for four (SYS <= 24V_GD limit <= 4.69 A). Start-up dissipation <= 6.7 W (resistive-load model, eq. 22-24) vs SLVSDG2G figure 55 at 85 C (graph reading, not verified). Asserted on every build. A shorted cell cable trips only its own eFuse. A PTC was rejected: MF-MSMF110 holds only 0.73 A at 60 C.

PWM LINE DRIVERS (EPWM13-16)



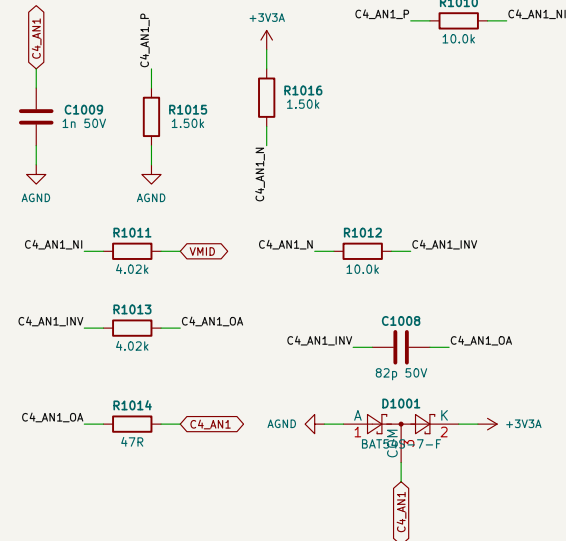
PWMs (1 = switch on) -> RS-422 pair, terminated on the power board. 10k pull-downs; ePWM balls are inputs without pull-up during reset/boot (table 6-8), so every gate command is 0 until firmware runs. G = C4_DRV_EN, G- high; outputs high impedance unless CHB_OK, C4_EN_MCU and RDY are high (INT-15). ePWM13-16 have no high-resolution PWM; buck-boost cells only, not a DAB bridge.

ANALOG FRONT ENDS AN1-AN3 (OPA4388), DUAL RANGE

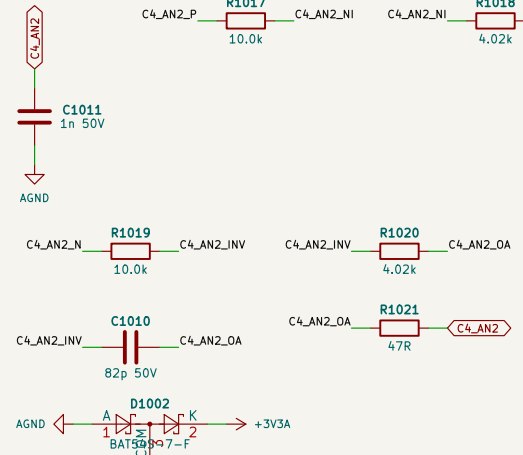


Difference amp per pair: $V_{\text{od}} = V_{\text{MID}} 1.25 \text{ V} + G \times (ANL_P - ANL_N)$, Rin 10.0k 0.1 X 10 ppm/K, $G = 4.02k/10.0k = 0.402$. Cf || Rf: anti-alias pole 483 kHz, 47 R + 1 nF COG; ADC charge bucket (ACQPS >= 320 ns at 12 bit). BAT5AS clamps the ADC node to 3V3A/AGND. All values calculated, not simulated. Accepts on any AN pair: differential on a 1.24-1.49 V common mode, legs 0.1-2.6 V (ADC span +/-3.1 V), or 0-3.0 V single-ended with ANL_N = AGND at the source (temperatures) -> 1.25-2.46 V. Op-amp inputs 0.9-1.75 V (RRIO). Source resistance adds to the 10.0k; keep R <= 100 R (1 % gain) or calibrate it out. AN1 = inductor current (INT-12, one constant); PVCELL-25 IEM LA 150-P 20.0 mV/A, 1.25 V CM, < 1 R source, linear +/-11.4 A -> 8.0k mV/A at the ADC. 0.0759 A/LSD; CMPS5 backup +/-91.5 A. The LA 150-P offset (up to 1.3 A) is re-zeroed by firmware at every idle (gates off); trip bands carry only the 1.17 A left after it. DAB-D60 (C1/C2): AN1 5.005 mV/A (+/-1.58 V max), AN2 <= 2.79 V, AN3 <= 2.73 V single-ended, 49.9 R sources (+0.5 % gain). Only AN1 carries the open-input bias. Channel D: VMID follower.

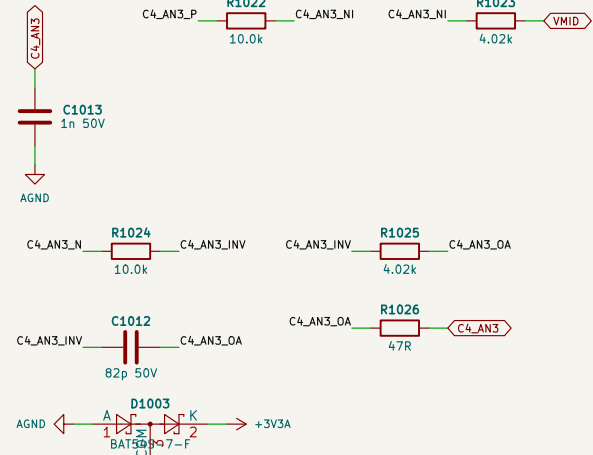
C4_AN1 -> ADCIND0 / CMPIN7P



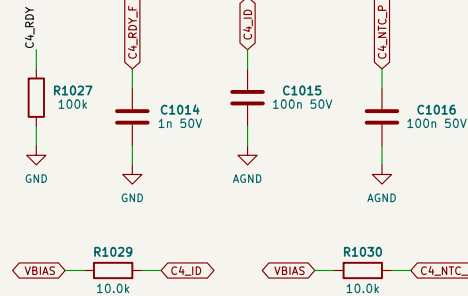
C4_AN2 -> ADCINA5



C4_AN3 -> ADCINB4



RDY, ID, NTC



RDY: push-pull from the power board (PVCELL-25 rev B; DAB-D60 rev B; LVC1G17, loff = high impedance when unpowered) or open drain with its own pull-up; the 100k pull-down here reads not-ready when the cable is unplugged or the board unpowered (INT-14). Levels asserted on every build: 1k/1 nF (1 us) filter into the GPIO and the driver-enable AND, ID: 10.0k from VBIAS (2.5 V), decode table in the pin plan (MUX_OUT). NTC (floating pair): 10.0k from VBIAS, 10k NTC reads 1.25 V at 25 C.

Not bench-validated. Values marked calculated are not simulated.
Sheet 11
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /10 Cell port C4/
File: 10_cell4.kicad_sch

Title: CTRL-C2000 control card - Cell port C4

Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 11/15

[illegible]

Pinout diagram for the TPS2600PWPR chip. The chip is shown as a yellow rectangle with pins numbered 1 through 17. Pin 1 is IN, Pin 2 is INL, Pin 3 is UVLO, Pin 4 is GVP, Pin 5 is MODE, Pin 6 is SHDN, Pin 7 is ILIM, Pin 8 is dVdT, Pin 9 is IMON, Pin 10 is FLT, Pin 11 is P_ILIM, Pin 12 is P_dVdT, Pin 13 is P_24V, Pin 14 is P_24V, Pin 15 is P_24V, Pin 16 is P_24V, Pin 17 is P_24V. The chip is labeled 'U1101' and 'TPS2600PWPR'. External components are shown: C1102 (1u 50V) connected to P_24V and GND, R1102 (121k) connected to P_ILIM and GND, C1103 (100n 50V) connected to P_dVdT and GND, and C1104 (1u 50V) connected to P_24V and GND.

U1102

Pin	Signal	Value	Connection
1	P_VA_IN1	3	+3V3A
2	P_VA_INV1	2	-INA
3	P_VAX_IN1	5	INB
4	P_VAX_INV1	6	-INB
5	P_VB_IN1	10	+INC
6	P_VB_INV1	9	-INC
7	P_VBX_IN1	13	-IND
8	P_VBX_INV1	12	-IND
9	P_VA_OUT1	1	+3V3A
10	P_VA_OUT2	2	+3V3A
11	P_VA_OUT3	3	+3V3A
12	P_VA_OUT4	4	+3V3A
13	P_VA_OUT5	5	+3V3A
14	P_VA_OUT6	6	+3V3A
15	P_VA_OUT7	7	+3V3A

U1103

Pin	Signal	Value	Connection
1	P_JA_IN1	3	+3V3A
2	P_JA_INV1	2	-INA
3	P_JB_IN1	5	INB
4	P_JB_INV1	6	-INB
5	P_AUX1_IN1	10	+INC
6	P_AUX1_INV1	9	-INC
7	P_AUX2_IN1	13	-IND
8	P_AUX2_INV1	12	-IND
9	P_JA_OUT1	1	+3V3A
10	P_JA_OUT2	2	+3V3A
11	P_JA_OUT3	3	+3V3A
12	P_JA_OUT4	4	+3V3A
13	P_JA_OUT5	5	+3V3A
14	P_JA_OUT6	6	+3V3A
15	P_JA_OUT7	7	+3V3A

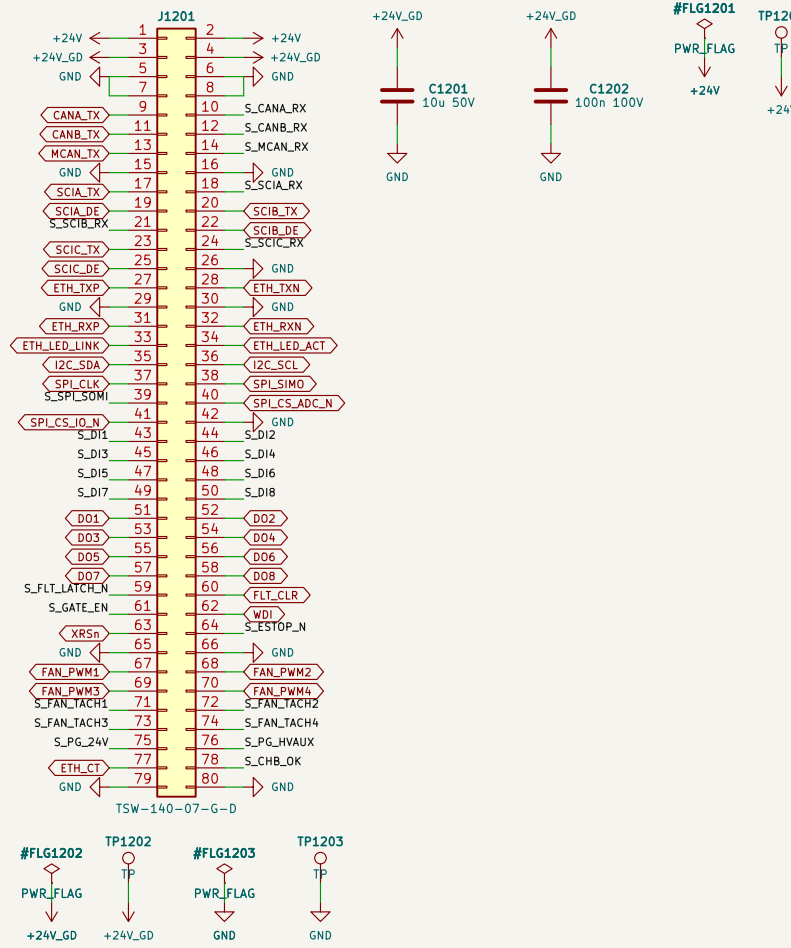
The schematic diagram illustrates the power supply section of the TMS320C6701 evaluation module. It features a +3V3A input connected to a network of resistors and capacitors. The network includes resistors R1122 (1.50k), R1123 (1.50k), R1118 (5.76k), R1119 (10.0k), R1120 (5.76k), and R1121 (47R). Capacitors C1112 (1n 50V) and C1111 (56p 50V) are also present. The network provides regulated outputs for P_VB_NI, P_VB_N, P_VB_INV, P_VB_OA, and P_VB. A D1103 Schottky diode is connected between the +3V3A input and the P_VB output line. The P_VB output is also connected to the BAT107 diode.

The schematic diagram illustrates the power supply section of the ADXL345 evaluation board. It features a +3V3A input connected to a network of resistors and capacitors. The circuit includes a 10.0k resistor network, a 50V capacitor, and a 56pF capacitor. The output is labeled P_VBX_M and P_VBX_N.

The schematic diagram illustrates the power supply circuit for the 68000 microprocessor. A +3V3A input is connected to a network of resistors and capacitors. The network consists of resistors R1139 (5.76k), R1140 (10.0k), R1141 (5.76k), R1142 (47R), R1143 (1.50k), and R1144 (1.50k). Capacitors C1118 (1n 50V) and C1117 (56p 50V) are also present. The circuit is grounded to AGND. The output of the network is connected to the VDD and VDDQ pins of the 68000 microprocessor.

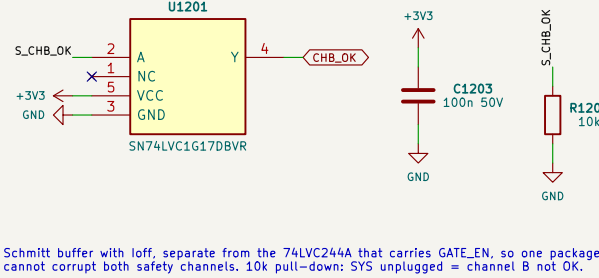
[illegible][illegible]

SYS CONNECTOR (SYS CONTRACT)



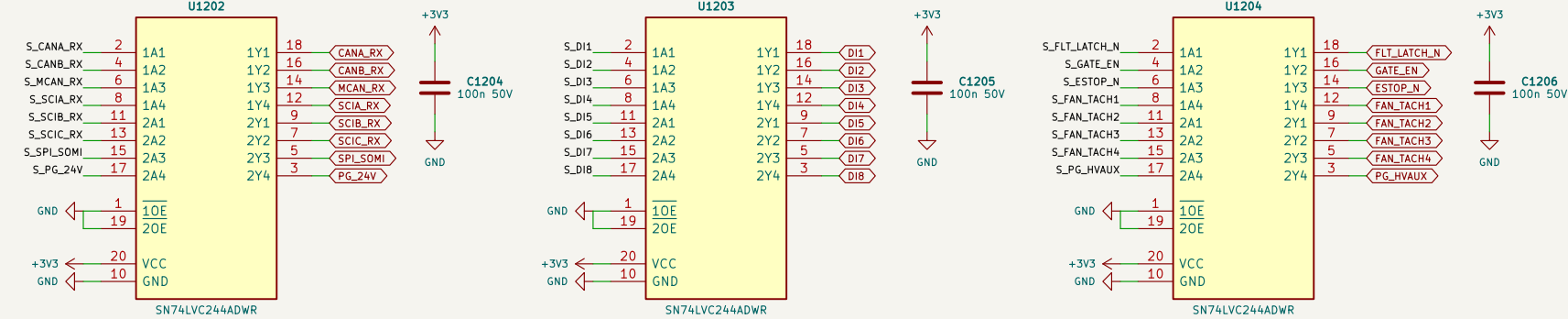
Pin map = Interfaces.SYS. SYS_RST_N lands directly on XRSn (open drain both sides).
+24V (pins 1-2): this card's regulators and the PORT connector. +24V_GD (pins 3-4): switched by safety channel B on SYS; goes ONLY to the four cell efuses (cell sheets). The two never meet here.
ETH_C0 (77): Port analog supply to the magnetics centre taps (Ethernet sheet).
Inputs from SYS arrive as S_<name> and reach the MCU as <name> through the buffers.

CHB_OK BUFFER (CHANNEL B, OWN PACKAGE)



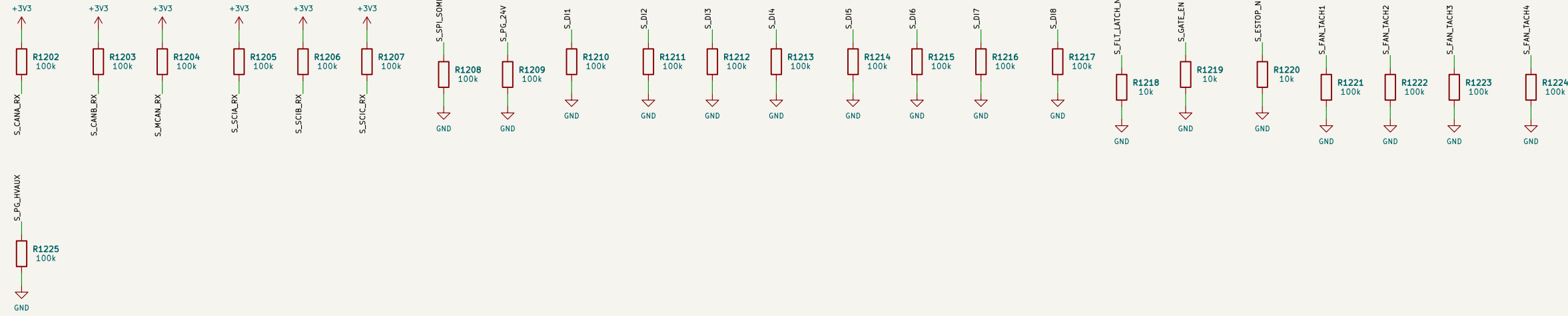
Schmitt buffer with ioff, separate from the 74LVC244A that carries GATE_EN, so one package cannot corrupt both safety channels. 10k pull-down: SYS unplugged = channel B not OK.

SYS INPUT BUFFERS (IOFF)



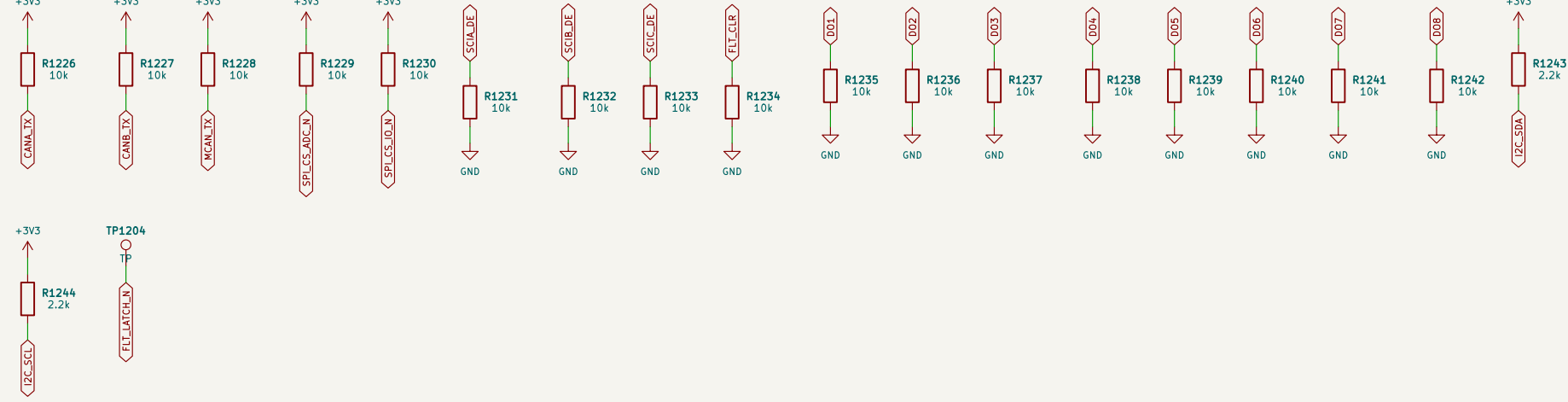
SYS makes its own 3.3 V and may drive these lines before this card's VDDIO is up: SPRSP14E 7.1 allows +/-20 mA total input-clamp current and 7.10.11-4.2 forbids driving pins early. 74LVC244A inputs and outputs are high impedance unpowered (ioff) and its outputs never exceed VDDIO. tpd ~5 ns.

INPUTS WITH SYS UNPLUGGED



FLT_LATCH_N, GATE_EN, ESTOP_N: 10k to GND, so an unplugged SYS reads as tripped and the gates stay off.
Other inputs: 100k to their idle level (CAN/SPI receive high, the rest low).

OUTPUTS WHILE THE MCU IS IN RESET



GPIOs float in reset; CAN TX recessive. RS-485 DE off. DO and FLT_CLR low. SPI chip selects high.
I2C: 2.2k pull-ups here (bus master side); SYS should not add a second pair.

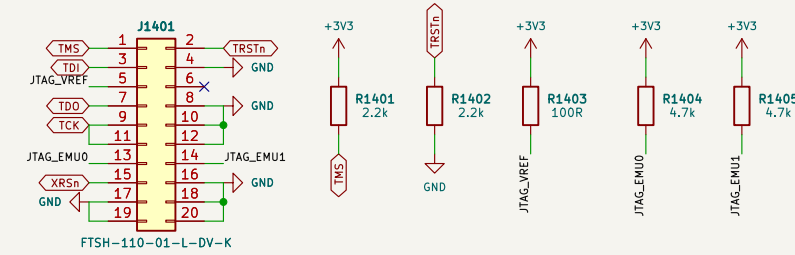
Not bench-validated. Values marked calculated are not simulated.
Sheet 13
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /12 SYS-IO-AUX connector/
File: 12_sys.kicad_sch

Title: CTRL-C2000 control card - SYS-IO-AUX connector

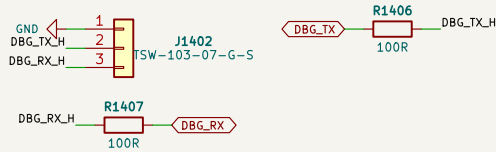
Size: A2 Date: 2026-10-04 Rev: E0
KiCad E.D.A. 10.0.6 Id: 13/15

JTAG CTI-20 (SPRU655I TABLE 13)



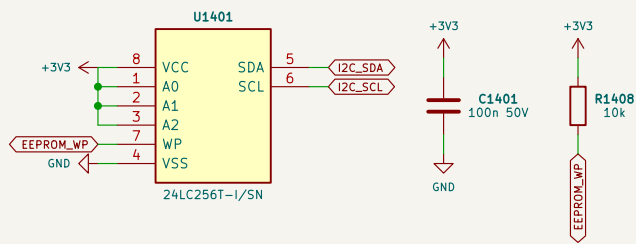
TMS 2.2k pull-up and TRSTn 2.2k pull-down (SPRSP14E table 6-1); TDIS to GND; VTRef via 100 R; RTCK looped to TCK; EMU0/1 4.7k pull-ups (no EMU pins on F2838x); RESET (open drain) on XRSn; EMU2-EMU4 (17-19) grounded as SPRSP14E 7.10.7 requires (CSR-07).

DEBUG UART (CM-UART-A, 3.3 V TTL)



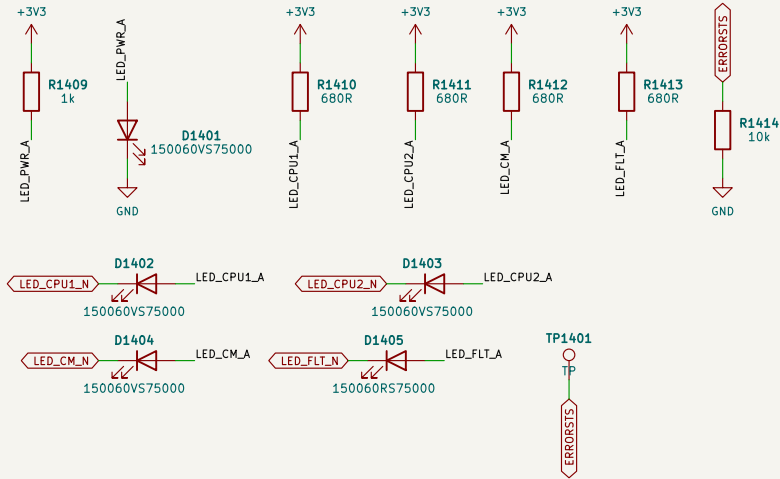
Pin 1 GND, 2 TX (from MCU), 3 RX (to MCU); 100 R series each way.

CALIBRATION / SERIAL-NUMBER EEPROM



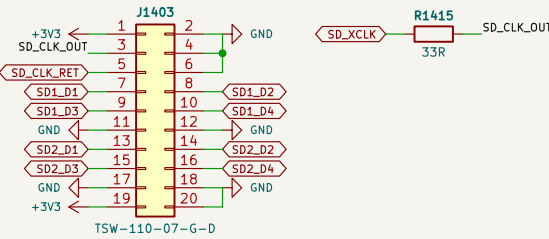
Address 1010111 (A2-A0 high = 0x57) so it cannot collide with a SYS EEPROM at the default 0x50. WP pulled high: write-protected unless firmware drives EEPROM_WP low.

STATUS LEDs AND ERRORSTS



GPIO sinks about 2 mA (680 R). ERRORSTS needs an external pull-down.

SDFM EXPANSION HEADER (NO MATING CONTRACT YET)



8 modulator data inputs (SD1/SD2 D1-D4) and one clock: XCLKOUT out, returned clock to all eight SDx_Cy inputs. 3.3 V CMOS; for an adjacent sensing board only.

Not bench-validated. Values marked calculated are not simulated.
Sheet 15
Schematics authored by Chinmoy Bhuyan
All interface contracts: gen/interfaces.py (CELL x4, PORT, SYS)

Sheet: /14 JTAG, UART, EEPROM, LEDs, SDFM/
File: 14_debug.kicad_sch

Title: CTRL-C2000 control card - JTAG, UART, EEPROM, LEDs, SDFM

Size: A3	Date: 2026-10-04	Rev: E0
KiCad E.D.A. 10.0.6		Id: 15/15